Isogeny-loop GLV, round 3: constructed curves, point-level validation, timings, binary GLS

2026-10-12 · study directory `research/isogeny_loop_glv_20261012/` · follows round 2 (`research/isogeny_loop_glv_20261011/`, PR #1698) and round 1 (`RESEARCH_ISOGENY_LOOP_GLV.md`, PR #1644) · status words: **measured**, **derived**, **proposed** · all computation native Rust (`isogeny_algos/`)

1. Summary

Round 2 left the 4-dimensional GLV-GLS split as a lead backed only by lattices and an operation-count model. On a quadratic GLS twist $E'/\mathbb{F}_{p^2}$, the split combines ψ ($\psi^2 = -1$) with a horizontal-isogeny loop endomorphism of a moderate-$|D|$ CM curve. This round builds the curves, checks the eigenvalues on points, times the split, and carries it to characteristic 2.

point-level validation	**measured, all pass.** Curves at $D \in \{-619, -23, -71, -431, -3019\}$, with p of 127 bits, are built from a natively computed H_D. ψ acts as $+\mu$, and each closing loop acts as $\pm\lambda(\pm s)$, on 24 points of order r (248–253 bits). The two closing walks land on conjugate embeddings.
benchmark, $D = -619$	**measured.** GLV-GLS-4 with a projective loop beats GLS-2 by 1.147× (run 1) and 1.091× (run 2, a different host), median/median. In run 1 it runs and 0.498× the time of wNAF; the A/A control reads 0.987. Across five D, the ratio is 1.06–1.21, which is 0.05–0.17 below round 2's model.
binary GLS	**constructed; validated at toy size.** Frobenius-assisted loops $\mathfrak{l} \cdot F^k$ exist on prime-order GLS twists at $m = 127$ and 131, but not at $m = 83$ for $l \le 61$. An x-only point check at $m = 31$ over GF(2^{62}) passes on 2 discriminants. **Not timed.**
native sweeper, x-only	**measured.** The port reproduces round 1 (12 curves; 911 + 911 + 306 scan discriminants) after a 131-bit defect was fixed. Under the x-only model no large-$|D|$ challenge curve has a non-Frobenius loop below the saving (upper bounds 5–13×).

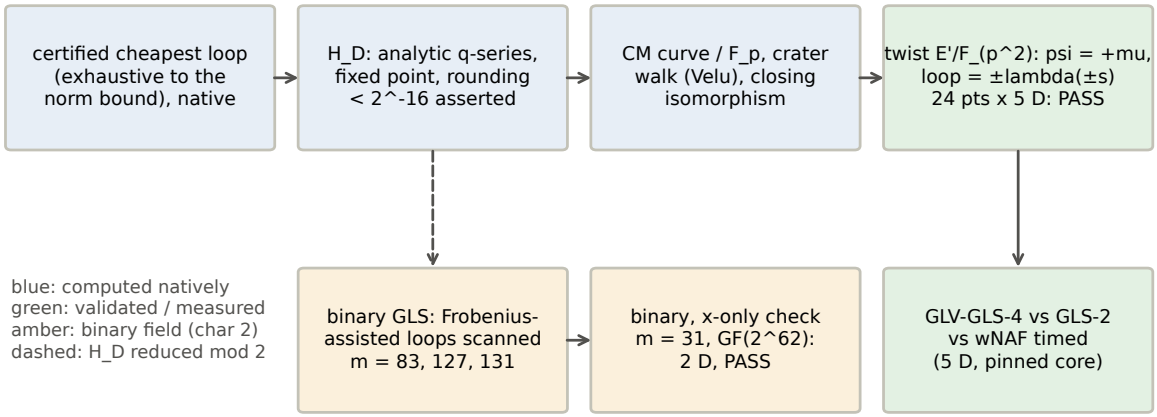


Figure 1: What round 3 computed natively (blue), validated or measured (green), and built in characteristic 2 (amber).

2. Point-level validation (measured)

`H_D` comes from `src/cm.rs`. It evaluates $j = E_4^3/\Delta$ on reduced forms in fixed point, with $1/q = e^{\pi\sqrt{|D|}/a}$ and the η-product computed by the pentagonal series. Precision grows until the worst coefficient rounding error is below 2^{-16}, and that bound is asserted. At this size the recorded bound is 2^{-86} to 2^{-118}.

A curve over $\mathbb{F}_p$ comes from a root of H_D, twisted if its trace is $-t$. A depth-first crater walk (`api::isogenies`, crater j only, no backtracking) follows the certified degree sequence back to j_0, and a closing isomorphism finishes the loop. Checks:

- on $E(\mathbb{F}_p)[r_E]$, each loop acts as one of $\pm\lambda(\pm s)$;
- on $E'(\mathbb{F}_{p^2})[r]$, $\psi = [\mu]$ with $\mu = (p-1)/t$;
- on $E'(\mathbb{F}_{p^2})[r]$, $\varphi\alpha\varphi^{-1} = [\pm\lambda(\pm s)]$, with $\lambda(s) = (A + Bs)/2$ and $s = (2\mu - t)/v$.

D	steps	H_D deg / bits / err	$\log_2 r$	ψ	twisted loops	verdict
−619	5·5·7	5 / 303 / 2^{-90}	251	$+\mu$ 24/24	$-\lambda(s)$; $-\lambda(-s)$	PASS
−23	2·3	3 / 150 / 2^{-86}	251	$+\mu$ 24/24	$-\lambda(-s)$; $-\lambda(s)$	PASS
−71	2·3·3	7 / 238 / 2^{-92}	248	$+\mu$ 24/24	$-\lambda(-s)$; $-\lambda(s)$	PASS
−431	2·2·3·3·3	21 / 631 / 2^{-118}	252	$+\mu$ 24/24	$+\lambda(-s)$; $+\lambda(s)$	PASS

−3019	5·5·5·13	7 / 523 / 2^{-88}	253	$+\mu$ 24/24	$-\lambda(-s)$; $+\lambda(s)$	PASS

Run `run-20261011T2330Z-gls-validate`. A replay on the final source reproduced the table byte for byte.

3. Timing (measured)

All arms run in one binary on the same arithmetic: `Fp2<FpM<2>>` (Montgomery, Karatsuba), a generic 127-bit p, Jacobian doubling and mixed addition, and $w = 5$ tables normalised by one batched inversion. Each GLV-GLS-4 call computes $\alpha'(P)$ and both ψ images. Every pair is checked before timing: all arms agree, $\psi(P) = [\mu]P$, $\alpha'(P) = [\lambda]P$, and each decomposition is $\equiv k$. The runs use 256 pairs and 30 alternating rounds, pinned to one core of a cloud Xeon at 2.10 GHz, not isolated.

arm, D = −619	median ns	min ns	wNAF / arm	GLS-2 / arm	note
wNAF	142,773	124,949	1.000	0.572	
GLS-2	81,665	71,901	1.748	1.000	125-bit coefficients
GLV-GLS-4, affine loop	73,066	67,012	1.954	1.118	63-bit coefficients
GLV-GLS-4, projective loop	71,172	64,788	2.006	1.147	
wNAF (A/A control)	144,657	125,440	0.987	—	noise floor

Stage medians: decomposition 2,850 ns; $\alpha'(P)$ 7,776 ns affine and 5,602 ns projective.

Confirmation run 2 (`run-20261012T0216Z-bench-confirm`): same command and seed, but a different host after a container restart (Xeon at 2.80 GHz, load average 1.23 at start). Medians: wNAF 211,895, GLS-2 114,823, GLV-GLS-4 105,273 (projective) and 106,268 (affine); A/A 0.989. That makes GLS-2 / GLV-GLS-4 = 1.091, against 1.147 in run 1. The 0.056 difference exceeds either run's A/A deviation; its cause is not isolated (host, load, and decomposition 2.9 vs 5.3 µs all changed).

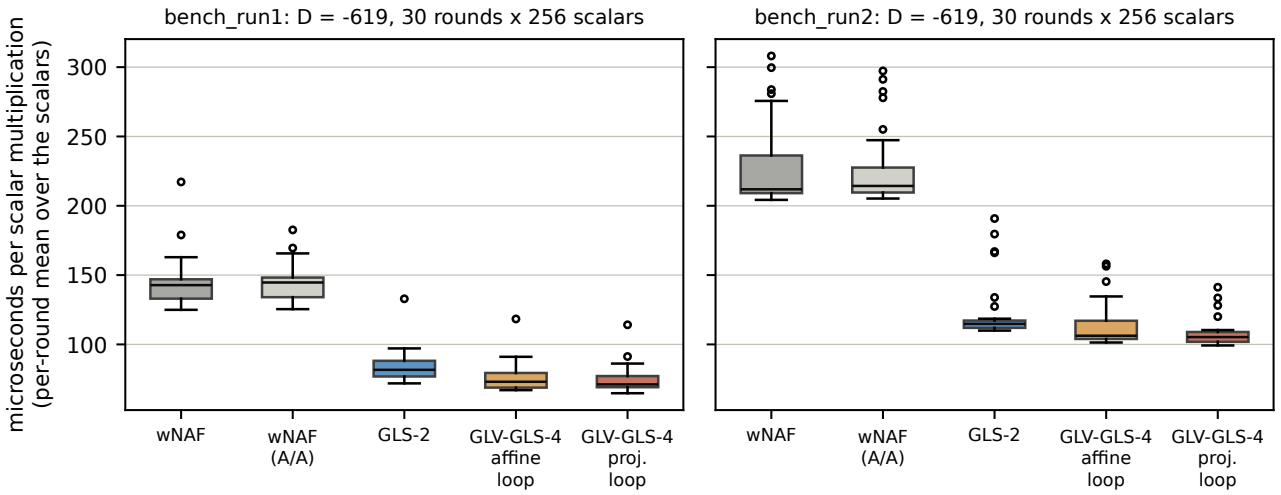


Figure 2: Per-round values (µs per scalar multiplication, mean over 256 scalars), 30 rounds, D = −619: run 1 (left, `run-20261011T2333Z-bench`) and run 2 (right, `run-20261012T0216Z-bench-confirm`, a different host).

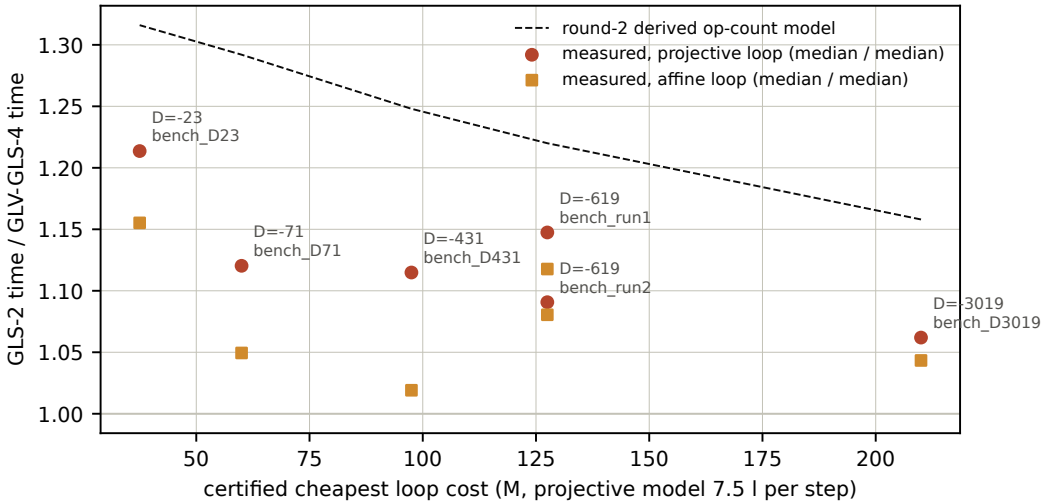


Figure 3: GLS-2 median / GLV-GLS-4 median at five D, against the certified loop's model cost, beside round 2's derived operation-count model. A/A controls 0.987–1.019.

Observations.

- The split was faster than GLS-2 at every D tested: 1.214, 1.120, 1.115, 1.147 (1.091 in run 2) and 1.062 with the projective loop.
- Every measurement is below the model. Two costs the model left out are visible in the stage timings:
  - the decomposition, about 3.5% of a GLS-2 call;
  - $\alpha'(P)$, 2–10.5% of a GLS-2 call. Its stage time rises with model cost: 1.7, 2.7, 4.2, 5.6 and 8.5 µs.
- The end-to-end ordering between D is not resolved below about 3%, because GLS-2 medians alone range 79.1–82.2 µs across runs.
- Scope: variable time, a generic p, one host, one build, no optimised-library comparison.

4. Binary-field GLS with Frobenius-assisted loops

Construction (derived).

- Take $E/\mathbb{F}_{2^m}$ ordinary, with $\pi = \left(t + v\sqrt{D}\right)/2$ and t odd. Then $D \equiv 1 \pmod 8$ and π generates $\mathfrak{p}_2^m$.
- An odd step $\mathfrak{l}$ followed by k squaring steps F closes when $[\mathfrak{l}][\mathfrak{p}_2]^k = 1$. Its degree is $l \cdot 2^{kk}$ with $kk = \min(k, m-k)$, and its cost is $c(l)$ plus squarings.
- Only F is cheap (round 2's erratum: V costs a doubling), so α must be F-oriented. A 2-adic root s_2 of D with $v_2(t + vs_2) > m$ fixes that direction, and α is F-oriented iff $v_2(A + Bs_2) > kk$.
- The quadratic twist over $\mathbb{F}_{2^{2m}}$ carries ψ as in the prime case, with $\#E' = (2^m - 1)^2 + t^2 = 2r$.

Scans (measured).

m	|D| range	scanned	ord = m	prime r	loop, l ≤ 61	cheapest (D: l, F steps, model M)
83	≤ 8·10⁶	810,576	440	4	0	—
127	≤ 4·10⁶	405,292	319	3	2	−1532327: 3, 53, 128.5 · −2854663: 43, 21, 364.5
131	≤ 8·10⁶	810,576	459	4	3	−283583: 13, 16, 129.5 · −359383: 7, 60, 172.5 · −1148567: 29, 59, 335.5

GLV-GLS-4 lattices give 64-bit coefficients ($m = 127$) and 66-bit ($m = 131$), against 126 and 130 for GLS-2 (5,000 scalars each). $m = 127$ is the field of GLS254 (Oliveira, López, Aranha, Rodríguez-Henríquez). $m = 131$ is ECC2K-130's base field, but these are moderate-D CM curves, not the $D = -7$ challenge curve.

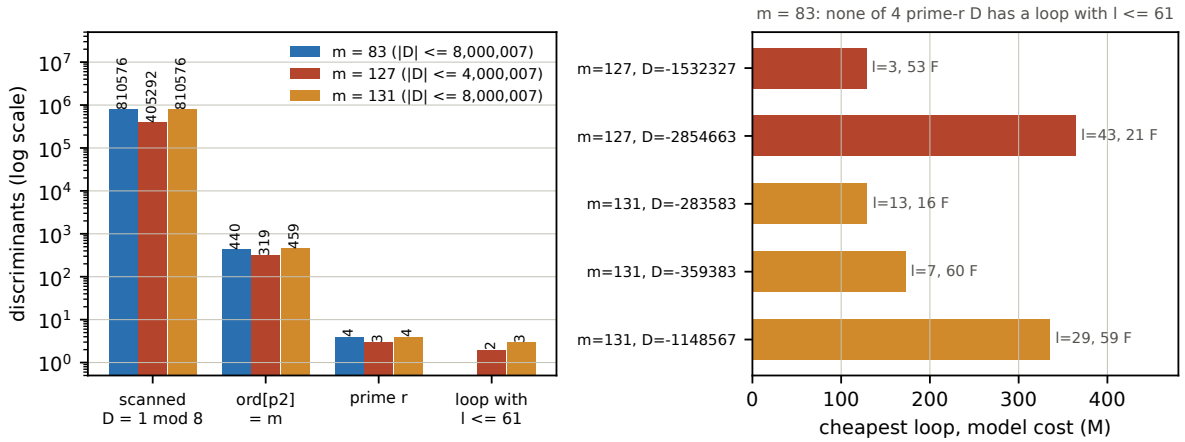


Figure 4: Funnel counts per m (log scale) and the cheapest loop's model cost per kept discriminant. Run `run-20261012T0000Z-binary-scan`.

Point-level validation at $m = 31$ (measured). At $m = 127$, $h \geq 127$, and the hit has $h = 1397$, so H_D is out of reach for this code. The check therefore runs at toy size: H_D mod 2, roots in GF(2^{62}), Kohel kernel polynomials for odd l, then F^{kk} back to j_0, and the twist $a_2 + \delta$ with $\mathrm{Tr}(\delta) = 1$. Twist and isomorphisms fix x in characteristic 2, so the checks are x-only:

- $x(\psi P) = x^q$ is compared with $x([\mu]P)$;
- $R(x)^{2^{kk}}$ is compared with $x([\lambda(\pm s)]P)$.

Results:

- $D = -39583$ ($h = 62$; $l = 31$, 11 F steps; H_D at 4,162 bits): PASS on 24 points.
- $D = -482623$ ($h = 186$; $l = 31$, 12 F steps; H_D at 14,081 bits): PASS on 24 points.

Run `run-20261012T0002Z-binary-validate`.

Cost model (derived, untimed).

- Over $\mathbb{F}_{2^{254}}$, 4-dimensional decomposition cuts the shared doublings from about 126 to about 64, the measured coefficient sizes. At about 4 M per doubling (López–Dahab or λ-coordinates, squarings free), that saves about 250 M.
- The $D = -1532327$ loop is one 3-isogeny (22.5 M) plus 53 F steps. The F steps compose to $x \mapsto x^{2^{53}}$, which a multi-squaring table applies in about 1–2 M.
- So the binary lead looks stronger than the prime one. In the prime case, though, the model overstated the measured gain by 0.05–0.17.
- Not examined: security beyond prime order, including GHS for $\mathbb{F}_{2^{2m}}$ with m prime and the moderate class number.

5. Native sweeper port and x-only re-run (measured)

Round 1's sweeper is now native: `examples/isogeny_loop_sweeper.rs` with a subexponential class group under GRH (`class_group_subexp.rs`: Bach-bound factor base, random-product relations with a large-prime variation, structured elimination, modular HNF and Smith form, discrete logs; verified by held-out relations, the analytic class number and brute force on all 6,079 fundamental $|D| < 20,000$) and the relation-lattice loop search (`loop_lattice.rs`: weighted LLL, exact DFS to a radius, verified generators, the 2-adic F/V orientation in characteristic 2). Three cost models share each class group: projective $7.5l$, affine $2l + 2$, x-only $2l$.

Reproduction of round 1. Groups a + d: h, cycle structure and cheapest projective cost equal PARI's on 12 of 13 curves (CryptoPro-A, 258 bits, skipped by both). Scans: 911 / 911 at 128 bits, 911 / 911 at 256 bits, 306 / 306 in characteristic 2 ($D \equiv 1$ mod 8), where the corrected Verschiebung price never changes the minimum because every cheapest loop there runs in the F direction. GROUPS_BC_PLACEHOLDER_TYP

A defect, found and fixed. The first port never finished a 131-bit discriminant: reduced forms have $a \leq \sqrt{|D|/3} \approx 2^{64.7}$ there, products of at most four factor-base primes have $a \leq 2^{62.3}$, so every trial product was already reduced and factored back into its own primes, a zero relation (104,276 of them hitting 1 of 2,547 columns in the diagnostic run). The base product is now built past `sqrt`$|D|$ by 12 bits, with a varying term count (a constant count trapped $D = -163$ at $h^* = 2$).

x-only verdicts. On the large-$|D|$ challenge curves of groups a + d the exhaustive search up to the saving finds no non-Frobenius loop in any model; the LLL upper bounds are 5.1–13.2× the saving under x-only (20–49× projective). Round 1's negative stands under the one model round 2's bound left open. The Koblitz, secp256k1 and CryptoPro-B loops are the known structure at 0.004–0.033 of the saving. In the scans every

small discriminant has a loop below the saving under x-only (median 0.078 at 128 bits, 0.039 at 256 bits): the constructive direction, not a statement about any given curve.

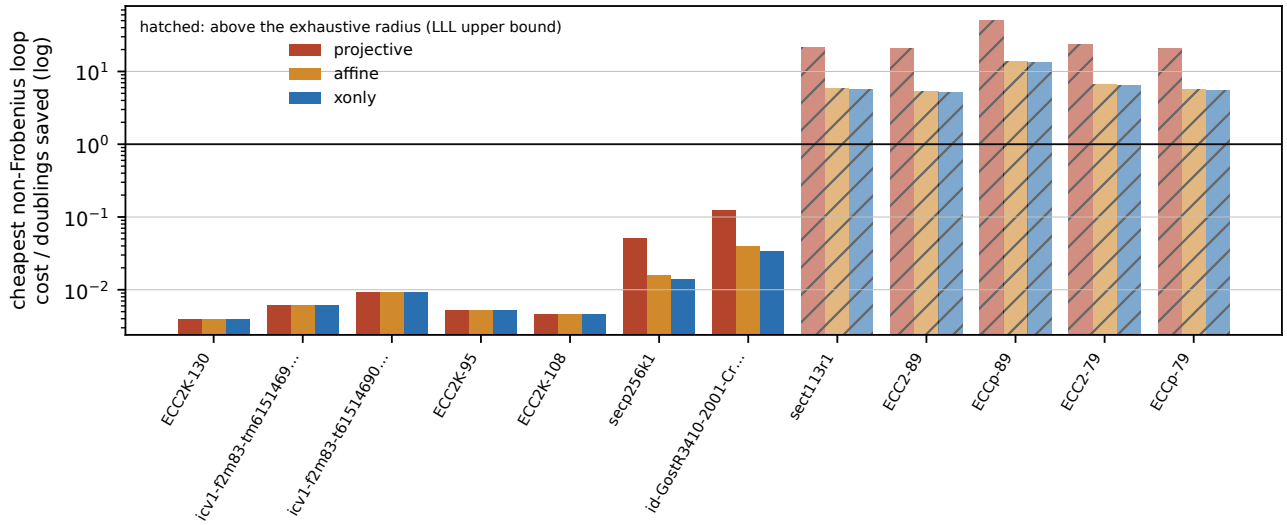


Figure 5: Cheapest non-Frobenius loop per challenge curve and cost model, as a fraction of the doublings saved; hatched bars are LLL upper bounds above the exhaustive radius. Run `run-20261018T1648Z-sweeper-native`.

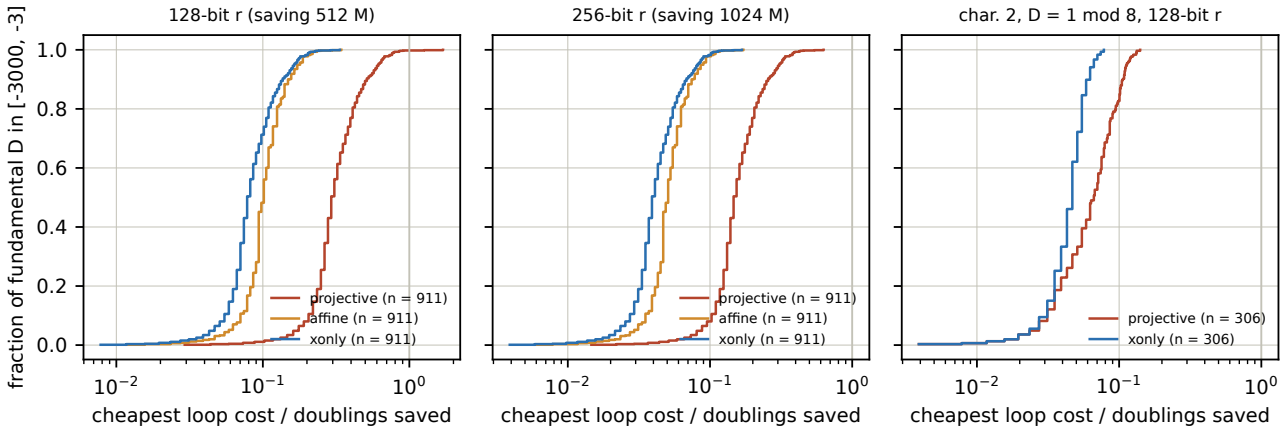


Figure 6: Cumulative distribution of the cheapest loop's cost / saving over the 911 scanned fundamental discriminants (306 in characteristic 2), per cost model.

6. Open after round 3

- A constant-time GLV-GLS-4 over a special prime. The condition is on D ($4p = t^2 + |D|\ v^2$), not on p, so Mersenne p is allowed. Does the shortfall against the model persist once the field is fast?
- A binary GLV-GLS-4 benchmark over $\mathbb{F}_{2^{254}}$ with carry-less multiplication.
- Groups b + c at 131 bits natively: a relation sieve and a sparse modular HNF (the class group is minutes to hours there against PARI's 2–6 minutes).
- Folding the x-only loop costs into the GLV-GLS model.

Evidence: `README.md` §7 indexes every record and run report.